

Beyond Degree: Rooted Motif Signatures for Latent Position Identifiability in Graphon Models

Roland B. Sogan

Sorbonne Université, Université Paris Cité, CNRS,
Laboratoire de Probabilités, Statistique et Modélisation,
roland-boniface.sogan@sorbonne-universite.fr

Tabea Rebafka

MIA AgroParisTech, INRAE, Université Paris-Saclay,
tabea.rebafka@agroparistech.fr

Abstract

Graphon estimation requires structural assumptions to address its intrinsic non-identifiability. A standard approach is degree-based identifiability, where the degree function is assumed to be strictly monotonic. This assumption is rather restrictive and fails for graphons with constant or non-injective degree function, even when distinct latent positions have different connectivity profiles. In this paper, we introduce *rooted motif signatures* as higher-order node-level representations for graphons. They extend the degree function by recording, at each latent position, the densities of rooted motifs such as triangles, cycles, paths, and other local subgraph patterns. We study the extent to which these signatures can distinguish latent positions beyond degree information. For generic finite-rank graphons, we prove that suitable rooted motif signatures determine the connectivity profiles of latent positions. We also explain why such a property cannot hold for arbitrary graphons without additional assumptions, since different latent positions may have identical rooted motif signatures. On the statistical side, we define empirical rooted motif signatures from a single observed graph and prove uniform concentration bounds for these estimators. Simulation experiments illustrate that rooted motif signatures can reveal latent structure in settings where degree-based representations are uninformative, including graphons with constant or non-injective degree functions and stochastic block models with equal block degrees.

1 Introduction

Graphons provide a fundamental framework for modeling dense random graphs. Originally introduced as limit objects for convergent sequences of dense graphs, they offer a nonparametric representation of connection probabilities between nodes and play a central role in network statistics, graph theory, and statistical learning (Lovász and Szegedy, 2006; Diaconis and Janson, 2007; Lovász, 2012). However, inferring a graphon from an observed graph raises fundamental difficulties, among which identifiability is one of the most prominent.

Indeed, in the graphon model, each node is associated with an unobserved latent position $U_i \in [0, 1]$, and the probability that two nodes i and j are connected is given by $W(U_i, U_j)$, where W denotes the graphon. This latent parametrization is not unique, since composing W with a measure-preserving transformation of $[0, 1]$ yields an equivalent graphon that defines the same probability distribution on graphs. Thus, graphons are naturally identifiable only up to measure-preserving equivalence. In many graphon estimation procedures, it is useful to select a canonical representative within this equivalence class. A standard way to do so is to construct a canonical representation of the latent positions. One common approach is based on the degree function

$$g(u) = \int_0^1 W(u, v) dv.$$

If g is injective, and in particular if it is strictly monotone, then expected degrees distinguish latent positions and induce a canonical ordering. The motivation behind degree-based assumptions is therefore not that they are necessary for graphon identifiability itself, but that they provide a simple route to latent-position identifiability, which in turn fixes a representative of the graphon within its equivalence class. Moreover, this ordering can be approximated from empirical degrees, which explains the role of degree-based assumptions in sorting-based and neighborhood-smoothing graphon estimators (Yang et al., 2014; Chan and Airolidi, 2014; Zhang et al., 2017; Sogan and Rebafka, 2026). Although successful in many situations, this degree-based strategy is rather restrictive. It fails whenever distinct latent positions have the same expected degree, even if they have different connectivity profiles. This situation occurs, for example, in stochastic block models with equal block degrees and in finite-rank graphons with constant or non-injective degree functions. To the best of our knowledge, there is no general framework for extending this form of latent-position identifiability beyond graphons with injective degree functions.

In this work, we introduce a new notion of identifiability based on *rooted motif signatures*. These signatures can be viewed as node-level analogues of the homomorphism and subgraph densities that characterize graphons at the global level (Lovász and Szegedy, 2006; Diaconis and Janson, 2007; Lovász, 2012). Instead of summarizing the whole graphon by global

motif densities, we attach to each latent position the collection of motif densities rooted at that point. The key idea is to enrich the local description of a node beyond its degree by incorporating higher-order rooted motif densities, such as rooted triangles, four-node motifs, cycles, and clique-like patterns. These quantities capture local connectivity patterns that are invisible to degree information alone and can therefore distinguish latent positions with identical expected degree but different connectivity profiles (Milo et al., 2002; Alon, 2007; Benson et al., 2016).

We investigate the extent to which rooted motif signatures determine latent connectivity profiles. In particular, we show that, for generic finite-rank graphons, suitable rooted motif signatures are rich enough to identify these profiles. We also explain why such identifiability cannot hold for arbitrary graphons without additional assumptions, since certain symmetries of the latent space preserve all rooted motif signatures. Finally, we develop the statistical counterpart of this construction. Given a single observed graph, we define empirical rooted motif signatures and prove uniform concentration bounds for these estimators. These bounds provide a statistical guarantee for comparing nodes through their motif signatures. Simulation experiments further illustrate that rooted motif signatures reveal latent structure beyond degree information, successfully separating nodes that remain indistinguishable under degree-based representations.

2 Preliminaries

2.1 Graphons

A graphon is a symmetric measurable function $W : [0, 1]^2 \rightarrow [0, 1]$, which can be interpreted as an edge-probability function for dense random graphs. To generate a random graph $G = (V, E)$ on n nodes, first sample latent variables $U_1, \dots, U_n$ independently and uniformly from $[0, 1]$. Conditionally on these latent variables, draw the entries of the adjacency matrix independently as

$$A_{ij}|U_i, U_j \sim \text{Bernoulli}(W(U_i, U_j)), \quad 1 \leq i < j \leq n,$$

set $A_{ji} = A_{ij}$, and put $A_{ii} = 0$. The edge set is then given by $E = \{\{i, j\} : A_{ij} = 1\}$. In this representation, the latent variable U_i plays the role of an unobserved node position, while the graphon W describes how connection probabilities vary across the latent space. Many classical random graph models can be expressed in this form. In particular, stochastic block models correspond to piecewise-constant graphons. Thus, graphons provide a natural nonparametric extension of block models and a flexible framework for statistical network

analysis (Diaconis and Janson, 2007; Lovász, 2012).

2.2 Identifiability of graphons

The identifiability problem in graphon models arises from the fact that the latent space is only defined up to measure-preserving transformations (Aldous, 1981; Hoover, 1979; Kallenberg, 2005; Diaconis and Janson, 2007). Let $\varphi : [0, 1] \rightarrow [0, 1]$ be a measure-preserving transformation, that is,

$$\lambda(\varphi^{-1}(A)) = \lambda(A) \quad \text{for every measurable set } A \subset [0, 1],$$

where λ denotes the Lebesgue measure. If $U \sim \text{Unif}[0, 1]$, then $\varphi(U) \sim \text{Unif}[0, 1]$. Consequently, the transformed graphon

$$W^\varphi(u, v) = W(\varphi(u), \varphi(v))$$

induces the same random graph model as W . This shows that graphons can be identified at most up to measure-preserving transformations. However, transformations of this form do not give the full equivalence relation between graphons. Indeed, two graphons may induce the same random graph distribution without being related by a single measure-preserving transformation. Diaconis and Janson (2007) give the example

$$W(u, v) = uv, \quad W'(u, v) = (2u \bmod 1)(2v \bmod 1).$$

These two graphons define the same exchangeable random graph model, but there is no measure-preserving transformation ψ such that $W(u, v) = W'(\psi(u), \psi(v))$ almost everywhere.

The appropriate statistical notion is therefore weak equivalence: two graphons are weakly equivalent when they induce the same random graph distribution. We write this equivalently as

$$\delta_\square(W, W') = 0,$$

where $\delta_\square$ denotes the cut distance (Lovász and Szegedy, 2006; Borgs et al., 2008; Diaconis and Janson, 2007). Recall that, for an integrable function U on $[0, 1]^2$, the cut norm is defined by

$$\|U\|_\square = \sup_{S, T \subseteq [0, 1]} \left| \int_{S \times T} U(u, v) \, du \, dv \right|,$$

where the supremum is taken over measurable sets S, T . The cut distance between two

graphons is then

$$\delta_{\square}(W, W') = \inf_{\varphi, \varphi'} \left\| W^{\varphi} - (W')^{\varphi'} \right\|_{\square},$$

where the infimum is taken over measure-preserving transformations $\varphi, \varphi' : [0, 1] \rightarrow [0, 1]$. The following result gives a characterization of this equivalence.

Theorem 2.1 (Diaconis and Janson (2007), Theorem 7.1). Let W and W' be two graphons. Then $\delta_{\square}(W, W') = 0$ if and only if there exist measure-preserving transformations $\varphi, \varphi' : [0, 1] \rightarrow [0, 1]$ such that

$$W(\varphi(u), \varphi(v)) = W'(\varphi'(u), \varphi'(v)), \quad \text{for almost every } (u, v) \in [0, 1]^2.$$

Thus, the observed graph distribution identifies only the weak equivalence class of W , not a unique representative of this class. A canonical representative can be obtained by following the approach of Borgs et al. (2010), which studies identifiability at the level of latent positions through their connectivity profiles. Two latent positions $u, v \in [0, 1]$ are called twins if $W(u, \cdot) = W(v, \cdot)$ almost everywhere. Twin latent positions have identical connectivity profiles. As a result, any statistic that depends only on the connection function $W(u, \cdot)$ assigns the same value to such positions. The twin relation therefore defines a quotient of the latent space. Writing

$$u \sim_{\text{twin}} v \iff W(u, \cdot) = W(v, \cdot) \text{ a.e.},$$

one obtains the quotient space $\Omega_W = [0, 1] / \sim_{\text{twin}}$. The elements of Ω_W are not necessarily intervals of $[0, 1]$. They are equivalence classes of latent positions having the same connectivity profile. In a stochastic block model, these classes correspond to the blocks, provided that distinct blocks have distinct connectivity profiles; blocks with identical rows in the block probability matrix are identified in the same quotient class. The Lebesgue measure induces a probability measure μ_W on Ω_W , and the quotient graphon is defined almost everywhere by $\tilde{W}([u], [v]) = W(u, v)$. Equivalently, if $q_W : [0, 1] \rightarrow \Omega_W$, $q_W(u) = [u]$ denotes the quotient map, then

$$W(u, v) = \tilde{W}(q_W(u), q_W(v)) \quad \text{for almost every } (u, v).$$

Thus, $\tilde{W}$ induces the same random graph distribution as W . It is therefore a representative of the same weak equivalence class, but defined on the quotient probability space (Ω_W, μ_W) rather than on the original interval. Results of Borgs et al. (2010) show that, within the class of twin-free representatives, weak equivalence coincides with isomorphism up to null sets. Consequently, twin-freeness removes the ambiguity created by duplicated latent positions;

it is therefore a necessary condition for any notion of latent-position identifiability based on connectivity profiles.

The quotient construction removes duplicated connectivity profiles, but it does not provide a scalar parametrization of the latent space. A classical way to obtain such a parametrization is to restrict attention to graphons whose latent positions can be distinguished by their expected degrees. This leads to the degree function

$$g(u) = \int_0^1 W(u, v) dv = \mathbb{E}[A_{ij} \mid U_i = u],$$

which is the first conditional moment of the connectivity profile of a node with latent position u . In particular, if the graphon admits a representative with a strictly monotone degree function, then expected degrees induce a canonical ordering of the latent space.

Condition 2.2 (Strict monotonicity of degree). A graphon W is said to satisfy the strict degree monotonicity condition if and only if there exists a representative W^{can} in the weak equivalence class of W such that

$$g^{\text{can}}(u) = \int_0^1 W^{\text{can}}(u, v) dv$$

is strictly increasing (or strictly decreasing) on $[0, 1]$.

Under this condition, latent positions are identifiable through their expected degrees, and the degree function provides a one-dimensional coordinate system on the latent space. This idea has been used in several graphon estimation methods (Chan and Airolidi, 2014; Yang et al., 2014; Olhede and Wolfe, 2014; Wolfe and Olhede, 2013; Sogan and Rebafka, 2026) and is explicitly presented as a way to define a canonical graphon representation (Yang et al., 2014; Chan and Airolidi, 2014).

However, strict monotonicity of degree is a restrictive assumption. It excludes graphons in which distinct latent positions have the same expected degree but different connectivity profiles. In such cases, the degree function cannot separate all latent positions, even though the corresponding graphon may still be twin-free. This limitation motivates replacing the scalar degree coordinate by richer local summaries of the connectivity profile.

3 Rooted local signatures and motif-based indistinguishability

The expected degree only captures first-order connectivity information. To obtain a richer description of a latent position u , we consider conditional subgraph densities rooted at u . These quantities generalize the degree function by measuring the expected occurrence of local motifs around a node located at u . They can be viewed as higher-order conditional moments of the connectivity profile $W(u, \cdot)$.

Definition 3.1 (Rooted motif density). Let (F, r) be a rooted graph, where $F = (V(F), E(F))$ is a finite simple graph and $r \in V(F)$ is a distinguished node called the root. For $u \in [0, 1]$, the rooted motif density of (F, r) at u is

$$t((F, r), W)(u) = \int_{[0,1]^{|V(F)|-1}} \prod_{\{a,b\} \in E(F)} W(u_a, u_b) \prod_{v \in V(F) \setminus \{r\}} du_v,$$

where $u_r = u$.

The quantity $t((F, r), W)(u)$ has a natural probabilistic interpretation. Fix the root node at latent position u , sample the remaining $|V(F)| - 1$ latent positions independently from the uniform distribution on $[0, 1]$, and generate edges according to the graphon W . Then $t((F, r), W)(u)$ is the probability that all edges of F are present among these nodes, with the root node fixed at u . It is therefore a conditional subgraph density attached to the latent position u . The degree function is recovered as the simplest rooted motif density. If $K_2^\bullet$ denotes the rooted edge, then

$$t(K_2^\bullet, W)(u) = \int_0^1 W(u, v) dv = g(u).$$

Similarly, if $P_3^\bullet$ denotes the path of length two rooted at one extremity, then

$$t(P_3^\bullet, W)(u) = \int_{[0,1]^2} W(u, v)W(v, w) dv dw.$$

This quantity measures the expected number of two-step connections emanating from a node located at u .

3.1 Rooted motif signatures

The rooted motif densities introduced above can be collected into a single descriptor of the latent position u . Since they summarize the occurrence of all finite rooted motifs around u , they provide a much richer characterization than the expected degree alone.

Let $\mathcal{F}_\infty$ denote the collection of all finite rooted motifs, considered up to rooted isomorphism.

Definition 3.2 (Rooted motif signature). The rooted motif signature of a latent position $u \in [0, 1]$ is the collection

$$\Phi_W(u) = (t((F, r), W)(u))_{(F, r) \in \mathcal{F}_\infty}.$$

The signature $\Phi_W(u)$ records all rooted motif densities around u . In particular, the degree function appears as the rooted edge motif, $t(K_2^\bullet, W)(u) = g(u)$. The signature therefore extends degree-based representations by incorporating higher-order local structure. Two latent positions having the same signature are indistinguishable through rooted motif statistics. This naturally induces a partition of the latent space into classes of motif-indistinguishable positions. In practice, one cannot estimate infinitely many motif densities. It is therefore natural to restrict attention to a finite collection of motifs. Let $\mathcal{F} = \{(F_1, r_1), \dots, (F_K, r_K)\}$ be a finite family of rooted motifs. The associated $\mathcal{F}$ -signature is

$$\Phi_W^\mathcal{F}(u) = (t((F_1, r_1), W)(u), \dots, t((F_K, r_K), W)(u)).$$

For instance, choosing

$$\mathcal{F} = \{K_2^\bullet, K_3^\bullet, P_3^\bullet, C_4^\bullet\},$$

where $K_2^\bullet$ is the rooted edge, $K_3^\bullet$ is the rooted triangle, $P_3^\bullet$ is the rooted path of length two, and $C_4^\bullet$ is the rooted four-cycle. The corresponding signature contains

$$\Phi_W^\mathcal{F}(u) = (g(u), g_\Delta(u), g_{P_3}(u), g_{C_4}(u)),$$

with, for example,

$$g_\Delta(u) = \int_{[0,1]^2} W(u, v)W(u, w)W(v, w) dv dw.$$

Definition 3.3 (Signature-based latent-position identifiability). A graphon W is said to be *signature-identifiable* if the rooted motif signature map $u \mapsto \Phi_W(u)$ is injective almost everywhere.

This definition is a latent-position identifiability condition within a given representative of the graphon. It says that two latent positions cannot have the same rooted motif signature, except possibly on a set of measure zero. Similarly, for a finite family $\mathcal{F}$ of rooted motifs, we say that W is $\mathcal{F}$ -signature identifiable if the finite-dimensional map $u \mapsto \Phi_W^{\mathcal{F}}(u)$ is injective almost everywhere. The degree function is recovered as the simplest rooted motif coordinate. Indeed, when $\mathcal{F} = \{K_2^\bullet\}$, $\mathcal{F}$ -signature identifiability coincides with almost-everywhere injectivity of the degree function. Thus, degree-based identifiability appears as the one-dimensional case of the rooted motif framework. More generally, if the degree function is injective and $K_2^\bullet \in \mathcal{F}$, then the signature map $u \mapsto \Phi_W^{\mathcal{F}}(u)$ is injective. The converse does not necessarily hold: rooted motif signatures may distinguish latent positions that have the same expected degree. The previous definition only requires separation of latent positions. It does not require the signature coordinates to induce an ordered representative. To formulate an analogue of strict degree monotonicity, one may impose the following stronger condition.

Definition 3.4 (Strictly ordered signature representation). Let $\mathcal{F}$ be a finite family of rooted motifs. We say that a graphon W admits a strictly $\mathcal{F}$ -signature ordered representation if there exists a representative W^{can} in the weak equivalence class of W such that, for almost every $u < v$,

$$t((F, r), W^{\text{can}})(u) \leq t((F, r), W^{\text{can}})(v) \quad \text{for all } (F, r) \in \mathcal{F},$$

and there exists at least one motif $(F, r) \in \mathcal{F}$ such that

$$t((F, r), W^{\text{can}})(u) < t((F, r), W^{\text{can}})(v).$$

This condition is stronger than $\mathcal{F}$ -signature identifiability. Signature identifiability only requires the map

$$u \mapsto \Phi_W^{\mathcal{F}}(u)$$

to separate latent positions almost everywhere. In contrast, the strictly ordered condition requires the existence of a representative in which all motif coordinates are nondecreasing along the latent order, with at least one coordinate strictly increasing for almost every pair $u < v$. Hence, not every $\mathcal{F}$ -signature identifiable graphon admits such a representation. The monotone representation should therefore be viewed as one possible way to select an ordered representative, analogous to strict degree monotonicity. Other choices are possible, for instance by ordering latent positions lexicographically according to their motif signatures. When $\mathcal{F} = \{K_2^\bullet\}$, this condition reduces to strict monotonicity of the degree function.

3.2 Finite-rank graphons

The previous definitions apply to arbitrary graphons. In general, however, it is not clear whether equality of rooted motif signatures implies equality of connectivity profiles. We therefore consider a class of graphons for which the latent structure can be analyzed explicitly. We now specialize to finite-rank graphons. Such graphons admit finite spectral representations through the associated graphon operator and have been studied in several contexts including graphon estimation, graphon centrality, and graphon control (Wolfe and Olhede, 2013; Avella-Medina et al., 2020; Gao and Caines, 2019). A graphon W is said to have finite rank m if the associated integral operator

$$(T_W f)(u) = \int_0^1 W(u, v) f(v) dv \quad (1)$$

has rank m . Equivalently, W admits a spectral decomposition

$$W(u, v) = \sum_{j=1}^m \lambda_j \phi_j(u) \phi_j(v), \quad (2)$$

where $\lambda_1, \dots, \lambda_m$ are the nonzero eigenvalues of T_W , and $\phi_1, \dots, \phi_m$ form an orthonormal family in $L^2([0, 1])$. For finite-rank graphons, the connectivity profile of a latent position u is completely determined by the spectral coordinates $\Theta(u) = (\phi_1(u), \dots, \phi_m(u))$. Indeed, $W(u, \cdot) = \sum_{j=1}^m \lambda_j \phi_j(u) \phi_j(\cdot)$, so that $W(u, \cdot) = W(v, \cdot)$ if and only if $\Theta(u) = \Theta(v)$.

Proposition 3.5. Let W be a finite-rank graphon of rank m . If $\lambda_i \neq \lambda_j$, ($i \neq j$), and $\langle 1, \phi_j \rangle \neq 0$, for $j = 1, \dots, m$, then the rooted path densities

$$t(P_k^\bullet, W)(u), \quad k = 2, \dots, m+1, \quad (3)$$

determine the spectral coordinates $\Theta(u)$.

Remark 3.6. Proposition 3.5 gives an explicit sufficient family of rooted motifs for generic finite-rank graphons. It shows that the rooted path densities $t(P_k^\bullet, W)(u)$, $k = 2, \dots, m+1$, determine the spectral coordinates $\Theta(u)$, and hence the connectivity profile $W(u, \cdot)$. Therefore, any rooted motif signature containing these path coordinates identifies twin classes in this setting. This result should not be interpreted as saying that rooted paths are the only informative motifs. They are used here because their relation to the spectral decomposition is explicit. Other rooted motifs may also distinguish latent positions, but the proposition only establishes the sufficiency of rooted paths.

A natural question is whether an analogous identifiability phenomenon can hold beyond

finite-rank graphons. More precisely, one may ask whether

$$\Phi_W(u) = \Phi_W(v) \implies W(u, \cdot) = W(v, \cdot) \quad \text{a.e.} \quad (4)$$

For arbitrary graphons, this implication is false. The obstruction comes from internal symmetries of the graphon. Suppose that there exists a measure-preserving transformation $\tau : [0, 1] \rightarrow [0, 1]$ such that $W(\tau(x), \tau(y)) = W(x, y)$ for a.e. $(x, y) \in [0, 1]^2$. Then, by applying the change of variables $z \mapsto \tau(z)$ in the definition of each rooted motif density, one obtains $t((F, r), W)(\tau(u)) = t((F, r), W)(u)$ for every rooted motif (F, r) and for almost every u . Hence $\Phi_W(\tau(u)) = \Phi_W(u)$ for a.e. u . This phenomenon occurs even for twin-free graphons. For instance, consider $W(u, v) = |u - v|$. The map $\tau(u) = 1 - u$ is measure-preserving and satisfies $W(\tau(u), \tau(v)) = W(u, v)$ for all $u, v \in [0, 1]$. Therefore,

$$\Phi_W(u) = \Phi_W(1 - u) \quad \text{for all } u \in [0, 1].$$

However, if $|u - \cdot| = |v - \cdot|$ a.e., then necessarily $u = v$, so W is twin-free. Thus twin-freeness alone does not imply signature-based latent-position identifiability. This example shows that the full rooted motif signature cannot, in general, identify latent positions beyond the symmetries of the graphon. This suggests the following natural question: if $\Phi_W(u) = \Phi_W(v)$, must u and v be related by a measure-preserving transformation τ satisfying $W(\tau(x), \tau(y)) = W(x, y)$ for a.e. (x, y) , with $\tau(u) = v$? Equivalently, do rooted motif signatures identify latent positions up to the internal measure-preserving symmetries of W ? We leave this question open.

4 Empirical rooted motif signatures

Degree-based graphon estimators use empirical degrees to recover a latent ordering under strict degree monotonicity (Yang et al., 2014; Chan and Airolidi, 2014). We extend this idea by replacing the scalar degree coordinate with a finite vector of empirical rooted motif densities, which can distinguish latent positions that have identical expected degree.

Let $\mathcal{F} = \{(F_1, r_1), \dots, (F_K, r_K)\}$ be a finite family of rooted motifs. Suppose that we observe the adjacency matrix $A = (A_{ij})_{1 \leq i, j \leq n}$ of an undirected graph generated from the graphon W . Throughout this section, the motif family $\mathcal{F}$ is fixed, and we assume that n is large enough so that $|V(F)| \leq n$, $(F, r) \in \mathcal{F}$. For each node i , define the empirical $\mathcal{F}$ -signature by

$$\widehat{\Phi}_i^{\mathcal{F}} = \left(\widehat{t}_i(F_1), \dots, \widehat{t}_i(F_K) \right), \quad (5)$$

where $\widehat{t}_i(F)$ denotes the empirical rooted motif density of F centered at node i . More precisely, if (F, r) is a rooted motif, then

$$\widehat{t}_i(F) = \frac{1}{(n-1)^{|V(F)|-1}} \sum_{\psi: V(F) \setminus \{r\} \hookrightarrow [n] \setminus \{i\}} \prod_{\{a,b\} \in E(F)} A_{\psi_i(a)\psi_i(b)}, \quad (6)$$

where $(n-1)_k = (n-1)(n-2) \cdots (n-k)$, the sum is over injective maps, and $\psi_i(r) = i$, $\psi_i(a) = \psi(a)$, for $a \neq r$. For example, the empirical rooted edge, triangle, path, and four-cycle densities are given by

$$\widehat{g}_i = \frac{1}{n-1} \sum_{j \neq i} A_{ij}, \quad (7)$$

$$\widehat{g}_{\Delta,i} = \frac{1}{(n-1)(n-2)} \sum_{\substack{j,k \neq i \\ j \neq k}} A_{ij} A_{ik} A_{jk}, \quad (8)$$

$$\widehat{g}_{P_3,i} = \frac{1}{(n-1)(n-2)} \sum_{\substack{j,k \neq i \\ j \neq k}} A_{ij} A_{jk}, \quad (9)$$

$$\widehat{g}_{C_4,i} = \frac{1}{(n-1)(n-2)(n-3)} \sum_{\substack{j,k,\ell \neq i \\ j,k,\ell \text{ distinct}}} A_{ij} A_{jk} A_{k\ell} A_{\ell i}. \quad (10)$$

Given nonnegative weights $w_1, \dots, w_K$, the empirical signatures induce the node-level dissimilarity

$$\widehat{d}_{\mathcal{F},w}(i, j) = \left(\sum_{k=1}^K w_k [\widehat{t}_i(F_k) - \widehat{t}_j(F_k)]^2 \right)^{1/2}. \quad (11)$$

At the population level, the corresponding pseudometric is

$$d_{\mathcal{F},w}(u, v) = \left(\sum_{k=1}^K w_k [t((F_k, r_k), W)(u) - t((F_k, r_k), W)(v)]^2 \right)^{1/2}. \quad (12)$$

The weights allow different motif coordinates to contribute at different scales. The unweighted Euclidean distance corresponds to $w_k = 1$ for all k . In practice, the weights may be chosen to standardize the motif coordinates, for instance using inverse empirical variances, or to penalize larger motifs. A simple size-based choice is $w_k = |E(F_k)|^{-\alpha}$, $\alpha \in [0, 1]$. When α is close to 0, all motifs contribute nearly equally. When α is close to 1, larger motifs receive smaller weights, which reduces the influence of more complex and typically more variable

motif counts. By construction,

$$d_{\mathcal{F}}(u, v) = 0 \iff \Phi_W^{\mathcal{F}}(u) = \Phi_W^{\mathcal{F}}(v).$$

Hence $d_{\mathcal{F}}$ becomes a metric on the quotient space induced by $\mathcal{F}$ -motif indistinguishability. Under $\mathcal{F}$ -signature identifiability, it defines a metric on the latent space up to null sets. This metric endows the latent space with a geometry induced by rooted motif statistics. Nodes that are close in this geometry have similar motif profiles and therefore similar local structural roles in the network. Such distances can be used to construct motif-based neighborhoods, perform clustering, learn low-dimensional representations of the latent space, and develop graphon estimation procedures.

The following result shows that empirical rooted motif signatures consistently estimate their population counterparts.

Theorem 4.1 (Uniform concentration of empirical rooted signatures). Let

$$\mathcal{F} = \{(F_1, r_1), \dots, (F_K, r_K)\}$$

be a fixed finite family of rooted motifs, and set $s_{\max} = \max_{1 \leq k \leq K} |V(F_k)|$. Assume that $n \geq s_{\max}$. Then there exists a constant $C > 0$, depending only on $\mathcal{F}$, such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_{1 \leq i \leq n} \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_2 \leq C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}. \quad (13)$$

In particular, if K is fixed, then

$$\max_{1 \leq i \leq n} \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_2 = O_{\mathbb{P}} \left(\sqrt{\frac{\log n}{n}} \right). \quad (14)$$

Theorem 4.1 shows that, uniformly over the observed nodes, empirical rooted motif signatures recover their population counterparts. Thus, for a fixed motif family $\mathcal{F}$, the vectors $\widehat{\Phi}_i^{\mathcal{F}}$ provide consistent data-driven coordinates for the latent motif representation. Since many downstream procedures depend on pairwise comparisons rather than on the individual signatures themselves, we next derive a uniform consistency result for the induced motif distances.

Corollary 4.2 (Uniform consistency of signature distances). Under the assumptions of Theorem 4.1, there exists a constant $C > 0$, depending only on $\mathcal{F}$, such that for every $\delta \in (0, 1)$,

with probability at least $1 - \delta$,

$$\max_{1 \leq i, j \leq n} \left| \widehat{d}_{\mathcal{F}, w}(i, j) - d_{\mathcal{F}, w}(U_i, U_j) \right| \leq C \left(\sum_{k=1}^K w_k \right)^{1/2} \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}.$$

In particular, if $\sum_{k=1}^K w_k = 1$, then

$$\max_{1 \leq i, j \leq n} \left| \widehat{d}_{\mathcal{F}, w}(i, j) - d_{\mathcal{F}, w}(U_i, U_j) \right| = O_{\mathbb{P}} \left(\sqrt{\frac{\log(nK)}{n}} \right).$$

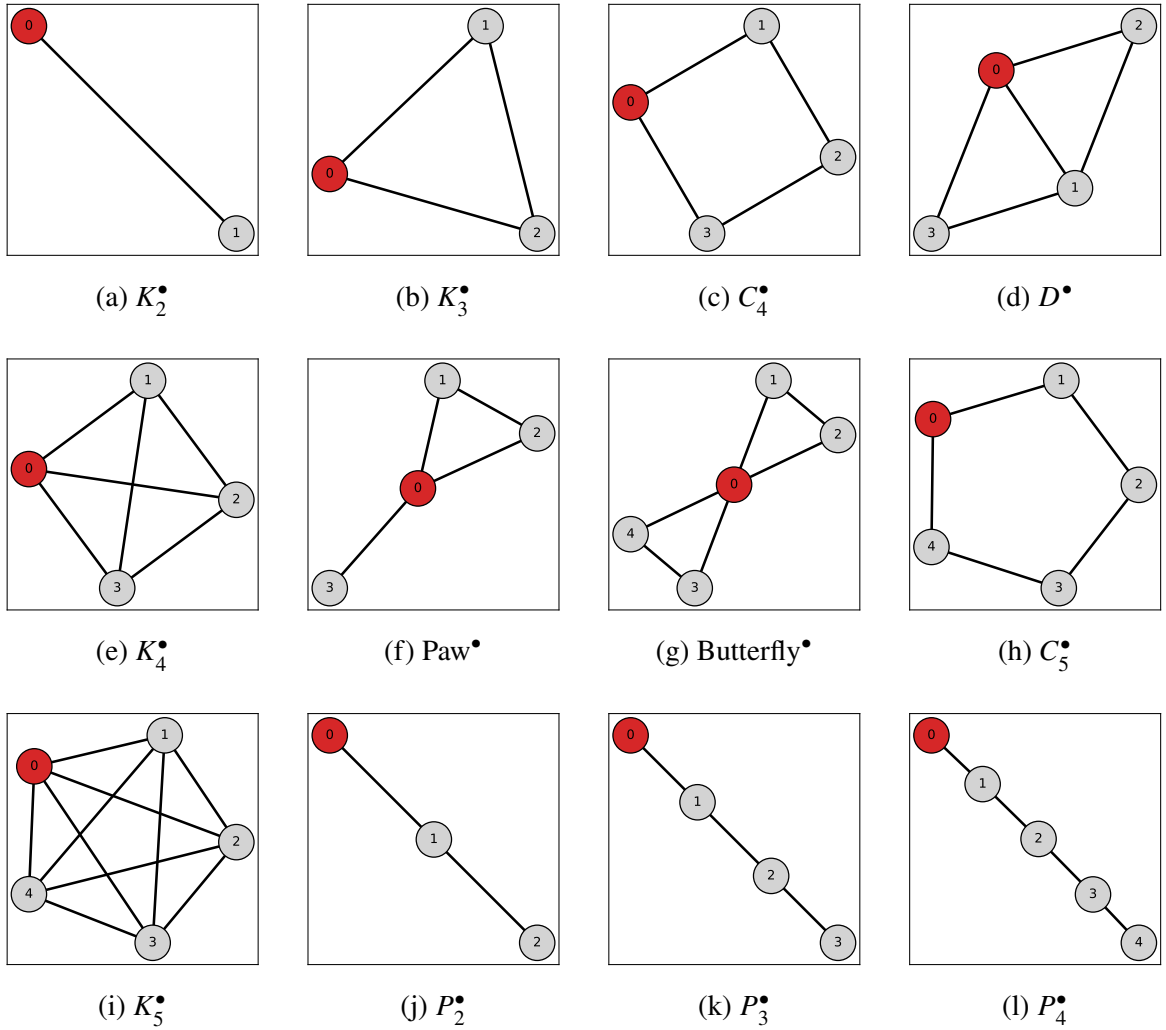


Figure 1: Rooted motifs used in the numerical experiments. The root vertex is highlighted in red.

5 Numerical experiments

In the numerical experiments, we do not restrict the signature to rooted paths. Although rooted paths are useful theoretically, in particular for the finite-rank result above, they may be highly correlated and therefore need not provide the most informative signature in practice. We instead use a collection of small rooted motifs designed to capture complementary local structures, including first-order and higher-order connectivity patterns. The selected rooted motifs are

$$\mathcal{F} = \{K_2^\bullet, K_3^\bullet, C_4^\bullet, D^\bullet, K_4^\bullet, \text{Paw}^\bullet, \text{Butterfly}^\bullet, C_5^\bullet, K_5^\bullet, P_2^\bullet, \dots, P_L^\bullet\}.$$

These are illustrated in Figure 1

5.1 Experiment 1: Stochastic block models

We first consider a collection of stochastic block models, a classical family of latent network models widely used for community detection and graphon approximation (Holland et al., 1983; Bickel and Chen, 2009; Abbe, 2018). The models are designed to test whether rooted motif signatures can distinguish latent classes beyond degree information. Each graph is generated from an stochastic block model with Q blocks, block proportions π , and connectivity matrix

$$P(A_{ij} = 1 \mid Z_i = a, Z_j = b) = B_{ab},$$

where $B \in [0, 1]^{Q \times Q}$ is symmetric and $Z_i \in \{1, \dots, Q\}$ denotes the latent block label of node i . We choose six stochastic block models, summarized in Table 1. The models are chosen to include assortative, disassortative, equal-degree, role-like, cyclic, and higher-dimensional block structures, which are common benchmarks for community detection and latent block modeling (Nowicki and Snijders, 2001; Rohe et al., 2011; Lei and Rinaldo, 2015).

For each model, we generate one graph with $n = 1000$ nodes and compute, for every node, its empirical rooted motif signature. The signature includes the empirical degree, rooted triangles, rooted four-cycles, rooted diamonds, rooted K_4 cliques, rooted paw motifs, rooted butterfly motifs, rooted five-cycles, rooted K_5 cliques, and walk-based rooted path coordinates. The resulting signature vectors are standardized and projected onto their first two principal components for visualization (Jolliffe, 2002). We compare this two-dimensional motif representation with the degree-only representation.

Figure 2 displays the results. In the left panel of each subfigure, nodes are represented only by their empirical degree, with a small vertical jitter added for visualization. In the right panel, nodes are represented by the first two principal components of their rooted

motif signatures. Colors indicate the true blocks, and ellipses summarize the dispersion of each block. The degree-only representation often produces substantial overlap between latent blocks, especially in the equal-degree setting. In contrast, the projection of rooted motif signatures onto the first two principal components provides a clearer separation of the latent blocks. This illustrates that higher-order rooted motif information captures structural differences between blocks that are not visible from degree information alone.

Table 1: SBM models used and connectivity matrices are in the Appendix 4.

Model	Q	π	Description
Assortative-2	2	(0.45, 0.55)	Two assortative blocks with unequal within-block strengths.
Disassortative-2	2	(0.40, 0.60)	Two disassortative blocks without exchange symmetry.
Equal-degree-3	3	(1/3, 1/3, 1/3)	Three balanced blocks with equal row sums but no block automorphism.
Role-3	3	(0.20, 0.45, 0.35)	Three asymmetric role-like blocks.
Broken-cycle-4	4	(0.18, 0.27, 0.25, 0.30)	Four-block cycle structure with broken rotational and reflection symmetries.
Gradient-5	5	(0.14, 0.19, 0.22, 0.25, 0.20)	Five-block gradient with unequal block masses.

5.2 Experiment 2: Continuous graphons with degree plateaux

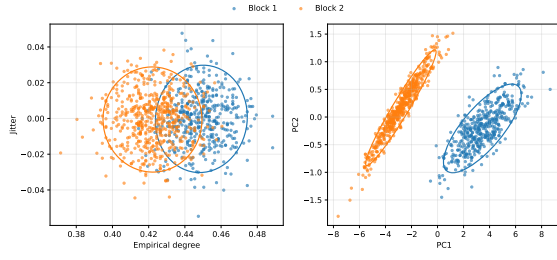
We next consider a continuous finite-rank graphon, a class closely related to spectral representations of graphon operators and low-rank network models (Wolfe and Olhede, 2013). We consider a graphon designed to exhibit plateau-like regions in the degree function while preserving non-trivial higher-order motif information, defined as

$$W_1(u, v) = 0.35 + 0.10 h(u)h(v) + 0.22 \sin(6\pi u) \sin(6\pi v), \quad (15)$$

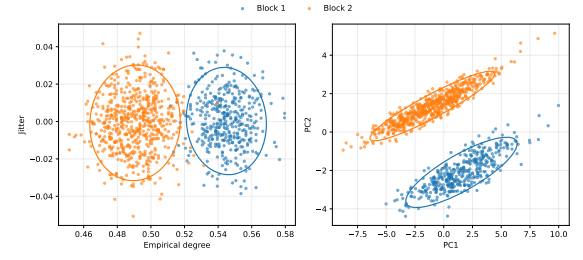
where

$$h(u) = s\left(\frac{u}{0.3}\right) \mathbf{1}_{\{u < 0.3\}} + \mathbf{1}_{\{0.3 \leq u \leq 0.7\}} + \left[1 + s\left(\frac{u - 0.7}{0.3}\right)\right] \mathbf{1}_{\{u > 0.7\}}, \quad s(x) = 3x^2 - 2x^3.$$

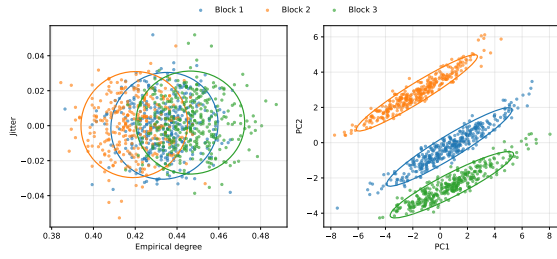
For this graphon, we compute the population rooted motif curves associated with the family of selected rooted motifs $\mathcal{F}$. These quantities are evaluated numerically through discretized graphon integral operators on a fine grid.



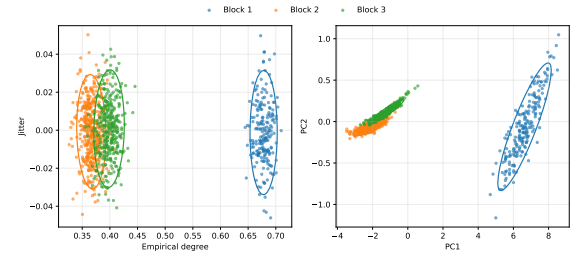
(a) Assortative-2



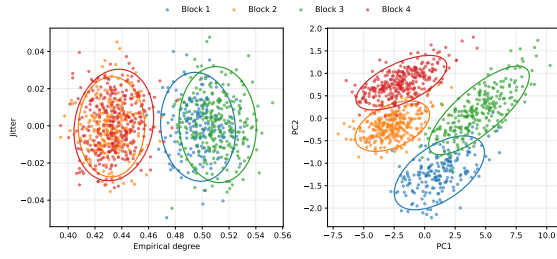
(b) Disassortative-2



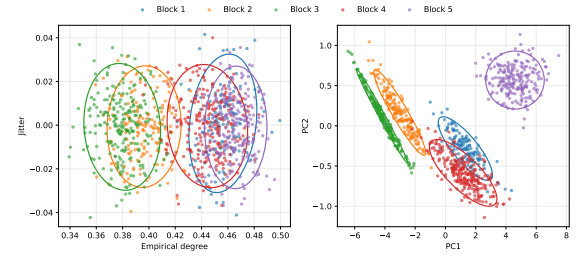
(c) Equal-degree-3



(d) Role-3



(e) Broken-cycle-4



(f) Gradient-5

Figure 2: Degree-only representation and PCA of rooted motif signatures for the SBM models.

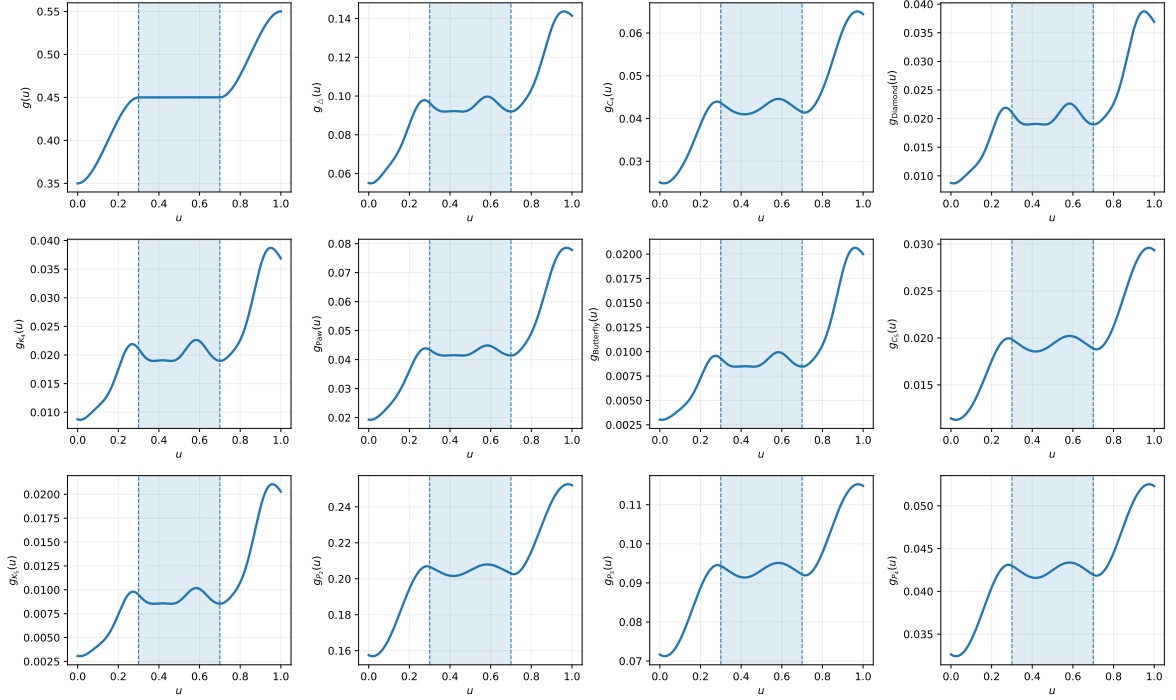


Figure 3: Population rooted motif curves associated with the graphon W_1 . The highlighted interval corresponds to the plateau region of the degree profile.

Figure 3 shows the resulting motif curves. The degree function exhibits extended regions where its variation is limited, suggesting that nodes located in those regions would be difficult to distinguish using degree information alone. In contrast, several higher-order motif coordinates vary substantially within the same intervals.

6 Discussion

This work proposes rooted motif signatures as a way to move beyond degree-based identifiability in graphon models. The main message is that local higher-order motif information can distinguish latent positions that degree alone cannot separate. In this sense, the identifiability based on rooted motif signatures provides a sufficient condition for constructing richer latent representations, while being less restrictive than strict degree monotonicity.

For finite-rank graphons, our results show that signature-identifiability coincides with the necessary graphon identifiability condition of Borgs et al. (2010). Indeed, under generic spectral assumptions, rooted motif signatures determine connectivity profiles and therefore separate all twin classes. The situation is more subtle for arbitrary graphons. Distinct latent positions may have identical rooted motif signatures even when their connectivity profiles are different. This can happen when the graphon is invariant under a nontrivial symmetry of

the latent space. In such cases, signature-identifiability cannot coincide with the necessary graphon identifiability condition in full generality.

Several extensions are natural. One direction is to use motif signatures for recovering latent groups, such as blocks in stochastic block models, especially in settings where degree information is uninformative. Another is to use motif-induced distances in graphon estimation procedures based on smoothing, nearest-neighbor averaging, or local aggregation. Rooted motif signatures could also be used as node features in downstream learning tasks, such as link prediction, node classification, or graph neural network architectures without external covariates. Finally, an important practical question is how to select the motif family and possibly learn motif weights, since different motifs may carry different information depending on the underlying graphon.

References

- Abbe, E. (2018). Community detection and stochastic block models: Recent developments. *Journal of Machine Learning Research*, 18(177):1–86.
- Aldous, D. (1981). Representations for partially exchangeable arrays of random variables. *Journal of Multivariate Analysis*, 11(4):581–598.
- Alon, U. (2007). Network motifs: Theory and experimental approaches. *Nature Reviews Genetics*, 8(6):450–461.
- Arcones, M. (1995). A bernstein-type inequality for u-statistics and u-processes. *Statistics & Probability Letters*, 22(3):239–247.
- Avella-Medina, M., Parise, F., Schaub, M. T., and Segarra, S. (2020). Centrality measures for graphons. *IEEE Transactions on Network Science and Engineering*, 7(1):319–335.
- Benson, A., Gleich, D., and Leskovec, J. (2016). Higher-order organization of complex networks. *Science*, 353(6295):163–166.
- Bickel, P. and Chen, A. (2009). A nonparametric view of network models and newman-girvan and other modularities. *Proc. Natl. Acad. Sci. USA*, 106(50):21068–21073.
- Borgs, C., Chayes, J., and Lovász, L. (2010). Moments of two-variable functions and the uniqueness of graph limits. *Geom. Funct. Anal.*, 19:1597–1619.
- Borgs, C., Chayes, J., Lovász, L., Sós, V., and Vesztegombi, K. (2008). Convergent sequences of dense graphs i: Subgraph frequencies, metric properties and testing. *Advances in Mathematics*, 219(6):1801–1851.

- Boucheron, S., Lugosi, G., and Massart, P. (2013). *Concentration Inequalities*. Oxford University Press.
- Chan, S. and Airoldi, E. (2014). A consistent histogram estimator for exchangeable graph models. In *International Conference on Machine Learning*, pages 208–216.
- Diaconis, P. and Janson, S. (2007). Graph limits and exchangeable random graphs. *Rendiconti di Matematica e delle sue Applicazioni, Series VII*, 28:33–61.
- Gao, S. and Caines, P. E. (2019). Spectral representations of graphons in very large network systems control. *IEEE Conference on Decision and Control*.
- Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30.
- Holland, P., Laskey, K. B., and Leinhardt, S. (1983). Stochastic blockmodels: First steps. *Social Networks*, 5(2):109–137.
- Hoover, D. N. (1979). Relations on probability spaces and arrays of random variables. Preprint, Institute for Advanced Study, Princeton, NJ.
- Jolliffe, I. T. (2002). *Principal Component Analysis*. Springer.
- Kallenberg, O. (2005). *Probabilistic Symmetries and Invariance Principles*. Springer, New York.
- Lei, J. and Rinaldo, A. (2015). Consistency of spectral clustering in stochastic block models. *The Annals of Statistics*, 43(1):215–237.
- Lovász, L. (2012). *Large Networks and Graph Limits*, volume 60 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI.
- Lovász, L. and Szegedy, B. (2006). Limits of dense graph sequences. *Journal of Combinatorial Theory, Series B*, 96:933–957.
- McDiarmid, C. (1989). On the method of bounded differences. In *Surveys in Combinatorics*, pages 148–188. Cambridge University Press.
- Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., and Alon, U. (2002). Network motifs: Simple building blocks of complex networks. *Science*, 298(5594):824–827.

- Nowicki, K. and Snijders, T. A. (2001). Estimation and prediction for stochastic blockstructures. *Journal of the American Statistical Association*, 96(455):1077–1087.
- Olhede, S. C. and Wolfe, P. J. (2014). Network histograms and universality of blockmodel approximation. *Proceedings of the National Academy of Sciences*, 111(41):14722–14727.
- Rohe, K., Chatterjee, S., and Yu, B. (2011). Spectral clustering and the high-dimensional stochastic blockmodel. *The Annals of Statistics*, 39(4):1878–1915.
- Sogan, R. B. and Rebafka, T. (2026). Low-complexity and consistent graphon estimation from multiple networks. In *Proceedings of The 29th International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research. PMLR.
- Wolfe, P. J. and Olhede, S. C. (2013). Nonparametric graphon estimation. *arXiv: Statistics Theory*.
- Yang, J., Han, C., and Airolidi, E. M. (2014). Nonparametric estimation and testing of exchangeable graph models. In *International Conference on Artificial Intelligence and Statistics*.
- Zhang, Y., Levina, E., and Zhu, J. (2017). Estimating network edge probabilities by neighborhood smoothing. *Biometrika*, 104(4):771–783.

A Proofs of theoretical results

A.1 Proof of Proposition 3.5

Proof. Since W has rank m , the associated integral operator satisfies

$$T_W f = \sum_{j=1}^m \lambda_j \langle f, \phi_j \rangle \phi_j.$$

Applying this identity to the constant function $\mathbf{1}$, we obtain

$$T_W \mathbf{1} = \sum_{j=1}^m \lambda_j \langle \mathbf{1}, \phi_j \rangle \phi_j.$$

Iterating the operator then gives, for every $k \geq 1$,

$$\begin{aligned}
T_W^k \mathbf{1} &= T_W^{k-1} \left(\sum_{j=1}^m \lambda_j \langle \mathbf{1}, \phi_j \rangle \phi_j \right) \\
&= \sum_{j=1}^m \lambda_j \langle \mathbf{1}, \phi_j \rangle T_W^{k-1} \phi_j \\
&= \sum_{j=1}^m \lambda_j \langle \mathbf{1}, \phi_j \rangle \lambda_j^{k-1} \phi_j \\
&= \sum_{j=1}^m \lambda_j^k \langle \mathbf{1}, \phi_j \rangle \phi_j.
\end{aligned}$$

Now observe that the rooted path density on $k + 1$ nodes is given by

$$\begin{aligned}
t(P_{k+1}^\bullet, W)(u) &= \int_{[0,1]^k} W(u, x_1) W(x_1, x_2) \cdots W(x_{k-1}, x_k) dx_1 \cdots dx_k \\
&= (T_W^k \mathbf{1})(u).
\end{aligned}$$

Hence, for $k = 1, \dots, m$,

$$t(P_{k+1}^\bullet, W)(u) = \sum_{j=1}^m \lambda_j^k \langle \mathbf{1}, \phi_j \rangle \phi_j(u).$$

Therefore the first m rooted path densities satisfy

$$\begin{pmatrix} t(P_2^\bullet, W)(u) \\ t(P_3^\bullet, W)(u) \\ \vdots \\ t(P_{m+1}^\bullet, W)(u) \end{pmatrix} = \begin{pmatrix} \lambda_1 c_1 & \lambda_2 c_2 & \cdots & \lambda_m c_m \\ \lambda_1^2 c_1 & \lambda_2^2 c_2 & \cdots & \lambda_m^2 c_m \\ \vdots & \vdots & & \vdots \\ \lambda_1^m c_1 & \lambda_2^m c_2 & \cdots & \lambda_m^m c_m \end{pmatrix} \begin{pmatrix} \phi_1(u) \\ \phi_2(u) \\ \vdots \\ \phi_m(u) \end{pmatrix},$$

where $c_j = \langle \mathbf{1}, \phi_j \rangle$. The matrix above can be written as

$$V \operatorname{diag}(c_1, \dots, c_m),$$

where

$$V = \begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_m \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_m^2 \\ \vdots & \vdots & & \vdots \\ \lambda_1^m & \lambda_2^m & \cdots & \lambda_m^m \end{pmatrix}$$

is a Vandermonde matrix. Since the eigenvalues $\lambda_1, \dots, \lambda_m$ are pairwise distinct,

$$\det(V) = \left(\prod_{j=1}^m \lambda_j \right) \prod_{1 \leq i < j \leq m} (\lambda_j - \lambda_i) \neq 0.$$

Therefore V is invertible. Moreover,

$$\det(\text{diag}(c_1, \dots, c_m)) = \prod_{j=1}^m c_j \neq 0$$

since $c_j = \langle \mathbf{1}, \phi_j \rangle \neq 0$ for every j . Hence $V \text{diag}(c_1, \dots, c_m)$ is invertible. Hence, the rooted path densities $t(P_k^\bullet, W)(u)$, $k = 2, \dots, m+1$, determine the spectral coordinates $\Theta(u)$. $\square$

A.2 Proof of Theorem 4.1

Proof. It is enough to prove the result for one fixed rooted motif (F, r) . Let

$$m = |V(F)|, \quad e = |E(F)|.$$

For a node i , write

$$\widehat{t}_i(F) = \frac{1}{(n-1)_{m-1}} \sum_{\psi: V(F) \setminus \{r\} \hookrightarrow [n] \setminus \{i\}} \prod_{\{a,b\} \in E(F)} A_{\psi_i(a)\psi_i(b)}.$$

Conditionally on $\mathcal{U} = (U_1, \dots, U_n)$, the conditional expectation of $\widehat{t}_i(F)$ is

$$\begin{aligned} \bar{t}_i(F) &:= \mathbb{E}[\widehat{t}_i(F) \mid \mathcal{U}] \\ &= \frac{1}{(n-1)_{m-1}} \sum_{\psi} \prod_{\{a,b\} \in E(F)} W(U_{\psi_i(a)}, U_{\psi_i(b)}). \end{aligned}$$

We decompose the error as

$$\widehat{t}_i(F) - t((F, r), W)(U_i) = \left(\widehat{t}_i(F) - \bar{t}_i(F) \right) + \left(\bar{t}_i(F) - t((F, r), W)(U_i) \right).$$

The first term is the edge-sampling error, conditionally on the latent variables $\mathcal{U}$. Changing an edge incident to the root i can affect at most $C_F n^{m-2}$ rooted embeddings, whereas changing an edge not incident to i can affect at most $C_F n^{m-3}$ rooted embeddings. Since the normalizing factor is of order n^{m-1} , the corresponding bounded-difference constants are respectively C_F/n

and C_F/n^2 . For an edge $e = (a, b)$, let

$$c_e = \sup_{A, A'} |\widehat{t}_i(F; A) - \widehat{t}_i(F; A')|,$$

where A and A' differ only in the entry corresponding to e . Therefore

$$\sum_e c_e^2 \leq n \left(\frac{C_F}{n} \right)^2 + n^2 \left(\frac{C_F}{n^2} \right)^2 \leq \frac{C'_F}{n}.$$

McDiarmid's inequality (McDiarmid, 1989) then yields

$$\mathbb{P}(|\widehat{t}_i(F) - \bar{t}_i(F)| > \eta | \mathcal{U}) \leq 2 \exp(-c_F n \eta^2). \quad (16)$$

We now control the second term, $\bar{t}_i(F) - t((F, r), W)(U_i)$. This is the latent-sampling error. Conditionally on U_i , the quantity $\bar{t}_i(F)$ is a bounded U -statistic of order $m - 1$, based on the latent variables $\{U_j : j \neq i\}$. Its kernel is

$$h_{F, U_i}(x_1, \dots, x_{m-1}) = \prod_{\{a, b\} \in E(F)} W(z_a, z_b),$$

where $z_r = U_i$ and $z_a = x_a$ for $a \neq r$. Since $0 \leq W \leq 1$, this kernel is bounded by 1. Moreover,

$$\begin{aligned} \mathbb{E}[\bar{t}_i(F) | U_i] &= \int_{[0,1]^{m-1}} \prod_{\{a, b\} \in E(F)} W(z_a, z_b) \prod_{a \neq r} dz_a \\ &= t((F, r), W)(U_i). \end{aligned}$$

Thus the second term is a centered bounded U -statistic. By the bounded-differences inequality for U -statistics (Boucheron et al., 2013; Hoeffding, 1963; Arcones, 1995), there exist constants $C_F, c_F > 0$, depending only on F , such that for every $\eta > 0$,

$$\mathbb{P}(|\bar{t}_i(F) - t((F, r), W)(U_i)| > \eta | U_i) \leq 2 \exp(-c_F n \eta^2). \quad (17)$$

Combining the edge-sampling bound 16 and the latent-sampling bound 17, we obtain, for every $\eta > 0$,

$$\mathbb{P}(|\widehat{t}_i(F) - t((F, r), W)(U_i)| > 2\eta) \leq 4 \exp(-c_F n \eta^2). \quad (18)$$

We now apply a union bound over all nodes and all motifs in $\mathcal{F} = \{(F_1, r_1), \dots, (F_K, r_K)\}$. Let

$$Z_{i,k} = |\widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i)|.$$

From (18), there exist constants $C_0, c_0 > 0$, depending only on $\mathcal{F}$, such that

$$\mathbb{P}(Z_{i,k} > s) \leq C_0 \exp(-c_0 n s^2)$$

for every $i = 1, \dots, n$, $k = 1, \dots, K$, and $s > 0$. Therefore,

$$\begin{aligned} \mathbb{P}\left(\max_{1 \leq i \leq n} \max_{1 \leq k \leq K} Z_{i,k} > s\right) &\leq \sum_{i=1}^n \sum_{k=1}^K \mathbb{P}(Z_{i,k} > s) \\ &\leq n K C_0 \exp(-c_0 n s^2). \end{aligned}$$

For every $\delta \in (0, 1)$, choosing $s = C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}$, we have

$$\mathbb{P}\left(\max_{1 \leq i \leq n} \max_{1 \leq k \leq K} |\widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i)| > C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}\right) \leq \delta.$$

Since

$$\left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_2 \leq \sqrt{K} \max_{1 \leq k \leq K} |\widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i)|,$$

it follows that, with probability at least $1 - \delta$,

$$\max_{1 \leq i \leq n} \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_2 \leq C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}.$$

This proves the theorem. □

A.3 Proof of the corollary 4.2

Proof. For $i, j \in \{1, \dots, n\}$, recall that

$$\widehat{d}_{\mathcal{F},w}(i, j) = \left\| \widehat{\Phi}_i^{\mathcal{F}} - \widehat{\Phi}_j^{\mathcal{F}} \right\|_w$$

and

$$d_{\mathcal{F},w}(U_i, U_j) = \left\| \Phi_W^{\mathcal{F}}(U_i) - \Phi_W^{\mathcal{F}}(U_j) \right\|_w,$$

where

$$\|x\|_w = \left(\sum_{k=1}^K w_k x_k^2 \right)^{1/2}.$$

Therefore,

$$\left| \widehat{d}_{\mathcal{F},w}(i, j) - d_{\mathcal{F},w}(U_i, U_j) \right| = \left| \left\| \widehat{\Phi}_i^{\mathcal{F}} - \widehat{\Phi}_j^{\mathcal{F}} \right\|_w - \left\| \Phi_W^{\mathcal{F}}(U_i) - \Phi_W^{\mathcal{F}}(U_j) \right\|_w \right|.$$

Using the reverse triangle inequality,

$$\left| \|a\|_w - \|b\|_w \right| \leq \|a - b\|_w,$$

with

$$a = \widehat{\Phi}_i^{\mathcal{F}} - \widehat{\Phi}_j^{\mathcal{F}}, \quad b = \Phi_W^{\mathcal{F}}(U_i) - \Phi_W^{\mathcal{F}}(U_j),$$

we obtain

$$\begin{aligned} \left| \widehat{d}_{\mathcal{F},w}(i, j) - d_{\mathcal{F},w}(U_i, U_j) \right| &\leq \left\| \left(\widehat{\Phi}_i^{\mathcal{F}} - \widehat{\Phi}_j^{\mathcal{F}} \right) - \left(\Phi_W^{\mathcal{F}}(U_i) - \Phi_W^{\mathcal{F}}(U_j) \right) \right\|_w \\ &= \left\| \left(\widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right) - \left(\widehat{\Phi}_j^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_j) \right) \right\|_w \\ &\leq \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_w + \left\| \widehat{\Phi}_j^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_j) \right\|_w. \end{aligned}$$

Taking the maximum over i and j , we get

$$\max_{1 \leq i, j \leq n} \left| \widehat{d}_{\mathcal{F},w}(i, j) - d_{\mathcal{F},w}(U_i, U_j) \right| \leq 2 \max_{1 \leq i \leq n} \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_w.$$

Moreover, for every i ,

$$\begin{aligned} \left\| \widehat{\Phi}_i^{\mathcal{F}} - \Phi_W^{\mathcal{F}}(U_i) \right\|_w &= \left(\sum_{k=1}^K w_k \left[\widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i) \right]^2 \right)^{1/2} \\ &\leq \left(\sum_{k=1}^K w_k \right)^{1/2} \max_{1 \leq k \leq K} \left| \widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i) \right|. \end{aligned}$$

By Theorem 4.1, with probability at least $1 - \delta$,

$$\max_{1 \leq i \leq n} \max_{1 \leq k \leq K} \left| \widehat{t}_i(F_k) - t((F_k, r_k), W)(U_i) \right| \leq C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}.$$

Hence, on the same event,

$$\max_{1 \leq i, j \leq n} \left| \widehat{d}_{\mathcal{F},w}(i, j) - d_{\mathcal{F},w}(U_i, U_j) \right| \leq 2C \left(\sum_{k=1}^K w_k \right)^{1/2} \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}.$$

In particular, if the weights are normalized so that

$$\sum_{k=1}^K w_k = 1,$$

then

$$\max_{1 \leq i, j \leq n} \left| \widehat{d}_{\mathcal{F}, w}(i, j) - d_{\mathcal{F}, w}(U_i, U_j) \right| \leq 2C \sqrt{\frac{\log(nK) + \log(1/\delta)}{n}}.$$

This proves the high-probability bound. $\square$

B Stochastic block model library

This appendix reports the stochastic block models used in Table 1. The corresponding connectivity matrices are displayed in Figure 4.

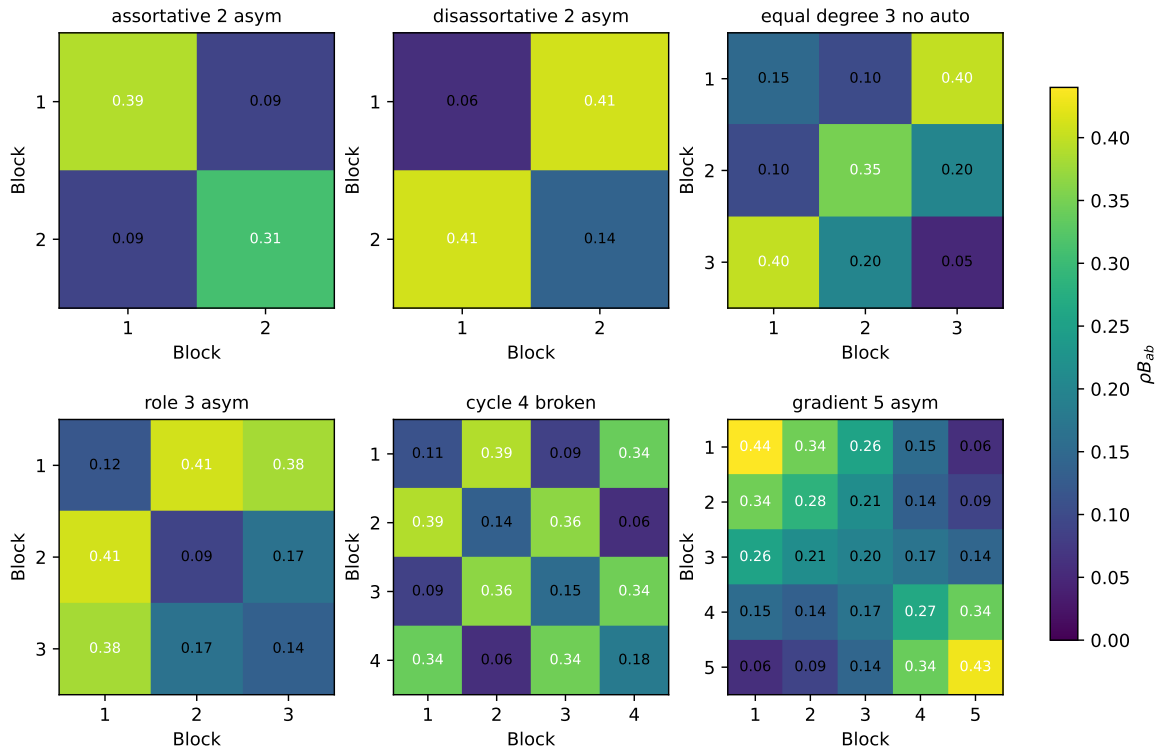


Figure 4: Connectivity matrices B defining the stochastic block model library used in Subsection 5.1.